

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

THURSDAY, 25 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are allowed to use a non-scientific calculator.

Answer questions only in the answer booklet provided.

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**SECTION A (Compulsory Questions)**

- Q1. (a) Based on the segment of codes below, write the statement in the cell with X to access the following element in array arr using subscript notation and offset notation with the specified pointers. Example on how to access element value 3 using ptr and ptr2 are given below. Redraw the table to show your answer. (8 marks)

```
int arr[10] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };
int *ptr;
int **ptr2;
int ***ptr3;

ptr = arr + 5;
ptr2 = &ptr;
ptr3 = &ptr2;
```

	Subscript Notation		Offset Notation	
Elements Value	3	8	3	8
Pointers				
ptr	ptr[-3]	X	*(ptr - 3)	X
ptr2	(*ptr2)[-3]	X	*(ptr2 - 3)	X
ptr3	X	X	X	X

- (b) Answer the following questions based on this program.

<pre>#include <iostream> using namespace std; struct Node { int item; Node* next; Node* bottom; Node(int newItem) { item = newItem; next = NULL; bottom = NULL; } }; int main () { int arr[3][3] = { {1, 2, 3}, {4, 5, 6}, {7, 8, 9} }; Node *temp[3][3]; Node *head; Node *cur, *cur2;</pre>	<pre>for (int i = 0; i < 3; i++){ for (int j = 0; j < 3; j++){ temp[i][j] = new Node(arr[i][j]); } head = temp[0][0]; for (int i = 0; i < 3; i++){ for (int j = 0; j < 3; j++){ if (j < 2) temp[i][j]->next = temp[i][j + 1]; if (i < 2) temp[i][j]->bottom = temp[i + 1][j]; } } return 0; } //end of main</pre>
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UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q1. (b)(Continued)**

- (i) Draw a linked list that is created from the program above (in the previous page). You need to show the item value, next and bottom pointer in each node and the head pointer in the list as well. (7 marks)
- (ii) Write a function void print(Node *head) to print all the nodes in the linked list created in Q1(b)(i). The head pointer is pointing to the first node in the list. A sample output of this function if it is used to print the list in Q1(b)(i) is shown below. (7 marks)

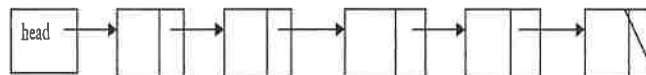
Sample Output:

1	2	3
4	5	6
7	8	9

- (iii) Briefly describe the effect of inserting a new node to the beginning and to the end of a general linked list. (3 marks)

[Total : 25 marks]

- Q2. (a) (i) Draw the content of the stacks s1 and s2 after the following segment of codes are executed according to the linked list format shown below in Figure 1.0. (10 marks)

**Figure 1.0 Example of Linked List Format**

```

Stack s1, s2;
char name[10][10] = { "Louis", "Rose", "Netty", "Sam", "Penny",
                      "Alice", "Kingston", "Betty", "Kenny",
                      "Mary" };
char c[5] = { 'S', 'B', 'A', 'K', 'N' };
char *str1, *str2;
bool found;

for (int x = 0; x < 10; x++) {
    found = false;
    for (int y = 0; y < 5; y++) {
        if (name[x][0] == c[y]) {
            s1.push(name[x]);
            found = true;
        }
    }
    if (!found)
        s2.push(name[x]);
}

```

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q2. (a)(Continued)**

- (ii) Draw the content of queue1 and queue2 after the following codes are executed. You must draw according to the linked list format shown in Figure 1.0. You must show the head and tail pointer as well. (10 marks)

```
int mysteryFunc(char *str){
    int i;

    for (i = 0; str[i] != '\0'; i++);
    return i;
}

int main(){
    Queue queue1, queue2;
    char name[10][10] = { "Louis", "Rose", "Netty", "Sam",
                          "Penny", "Alice", "Kingston", "Betty",
                          "Kenny", "Mary" };
    char *temp1 = new char[10];
    int val;

    for (int i = 0; i < 10; i++) {
        val = mysteryFunc(name[i]);
        if (val % 2 == 0) {
            queue1.enqueue(name[i]);
        }
        else
            queue2.enqueue(name[i]);
    }
    return 0;
}
```

- (b) Write a function **void remove(char *str)** to remove space, full stops and coma from the string using **Queue**. An example for the content in string str before and after the function call is shown below:

str Before Function Call:

Queue is used to simulate scenario in bank, supermarket, print queue and etc..

str After Function Call:

Queue is used to simulate scenario in bank, supermarket, print queue and etc..

(5 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

- Q3. (a) (i) The integer numbers in Figure 2.0 below will be sorted using Quick Sort in ascending order. The first partition of the list and the pivot value is shown in the shaded box. Redraw the table in Figure 2.0 by filling up the boxes with "X" label and fill in the index value at the last column.

(9 marks)

63 13 86 41 8 33 77

63	13	86	41	8	33	77	index = 0
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
41	X	X	X	X	X	X	index = ?
X	X	X	41	X	X	X	index = ?

Figure 2.0: First Partition in a Quick Sort

- (ii) Provide and explain a possible worst case scenario that could happen in sorting using Quick Sort method. (2 marks)
- (b) Search for character 't' in the array below using binary search method and provide the values for lower, mid and upper for every search step in a table.

'a'	'c'	'e'	'h'	'k'	'o'	'r'	't'
------------	------------	------------	------------	------------	------------	------------	------------

(4 marks)

- (c) Table below shows the operations that applied on sorted circular doubly linked list and unsorted linked list. Assume that all items are sorted in ascending order, identify the complexity in Big-O notation for the worst case scenario in the table below.

Operation	Sorted Circular Doubly Linked List	Unsorted Linked List
Finding largest element		
Deleting smallest element		
Looking for a specified element		

(6 marks)

- (d) Compare the complexity for the best and worst case scenario when perform searching using Binary Search and search in a Hash Table. (4 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Section B (Choose Any One Question Only)**

- Q4. (a) Draw the Binary Search Tree (BST) if the following values are inserted into a tree from left to right one at a time. The tree is empty before the first value is inserted.

9 5 13 2 3 11 16 15 88

(4 marks)

- (b) The following questions (b(i) and b(ii)) are based on the Binary Search Tree (BST) drawn in Q4 (a).

(i) Redraw the tree after node 9 is deleted. (3 marks)

(ii) State whether the Binary Search Tree in Q4 (a) (before deletion) is an AVL tree? If not, specify which node is not balanced and identify the unbalance pattern (eg. LL, LR, LB, RR, RB, RL). Draw the tree after the balancing. (5 marks)

- (c) Assume that we have numbers between 1 and 100 in a Binary Search Tree (BST) and want to search for the number 50. State whether the examination of the following sequences for the nodes is valid or not in order to find 50 and justify your answer.

(i) 55, 56, 57, 58, 59 (2 marks)

(ii) 55, 43, 47, 49, 48 (2 marks)

(iii) 100, 99, 98, 50, 49 (2 marks)

- (d) Write a recursive function **int TreeHeight(BTNode *nodePtr)** to return the height of a Binary Search Tree (BST). (7 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

Q5. (a) Answer question 5(a) using the Undirected Graph A shown in Figure 3.0.

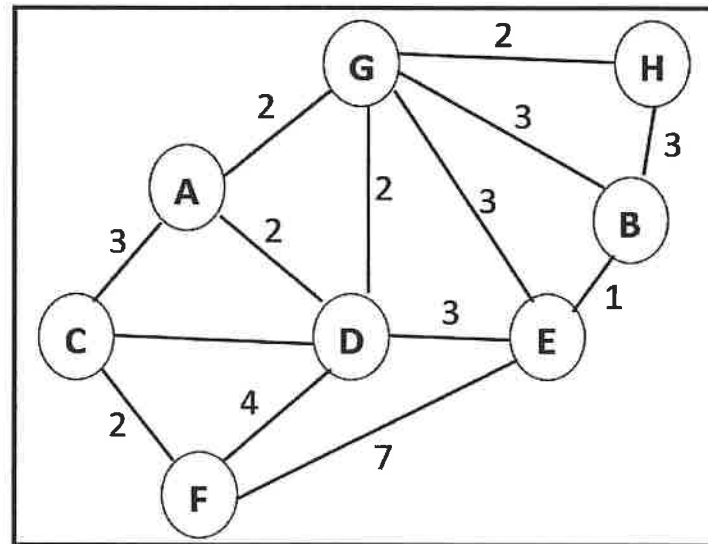


Figure 3.0: Undirected Graph A

- (i) Construct an adjacency matrix representation table based on the Undirected Graph A as shown in Figure 3.0. (4 marks)
 - (ii) The vertices are traversed alphabetically. Starting from vertex A, traverse the graph using Depth-First-Search (DFS) and Breadth-First Search (BFS). Provide the vertices order for each traverse. (8 marks)
 - (iii) Assume the graph in Figure 3.0 represents a map of a town and every vertex represents a house in the city. The edge is used to connect the houses and will be assigned a value to represent the distance between 2 houses. Suggest a method to help a telephone company to wire up all the houses with phone line by using the least amount of telephone wiring possible. (3 marks)
- (b) Answer the following questions based on the recursive function defined in Figure 3.1.

```

int Func(int n) {
    if(n <= 1)
        return n;
    else
        return Func(n - 1) + Func(n - 2);
}

```

Figure 3.1 Recursive Function

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q5. (b)(Continued)**

- (i) Describe the operation performed in this function Func. (2 marks)
- (ii) Convert Func into a non-recursive function. (4 marks)
- (iii) If compare recursive function defined in Figure 3.1 with the non-recursive function in Q5(b)(ii), which one is more efficient? Justify your answer. (4 marks)

[Total : 25 marks]